Ch. 1 - Speaking Mathematically

Set Roster Notation

$\{1, 2, 3\}$

$\mathbb{R} \Rightarrow$ set of all real numbers
$\mathbb{Z} \Rightarrow$ set of all integers
$\mathbb{Q} \Rightarrow$ set of all rational numbers
$\mathbb{N} \Rightarrow$ set of all natural numbers

Graph $= G(V, E)$ Edges
vertices

Degree of v = $\deg(v)$

Set Builder Notation

$\{x \in S \mid P(x)\}$

set of all such that

Ordered Pair

(a, b)

Subset

$A \subseteq B$

proper subset

$A \subset B$

Cartesian Product

$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}$

Relation R from A to B $\Rightarrow$ $A \times B$

domain codomain

$x R y \Rightarrow (x, y) \in R$

$x \not R y \Rightarrow (x, y) \notin R$

Arrow Diagram

if, and only if, xRy
if, and only if, $(x, y) \in R$

Function Notation

$F(x)$ F of x

f equals g if $f = g$,
if and only if,
$f(x) = g(x)$ for all x in A

Ch. 2 - Logic of Compound Statements

Negation $\neg p$

Conjunction $p \wedge q$

Disjunction $p \vee q$

Logically Equivalent

$P \equiv Q$

De Morgan's Law

$\neg(p \wedge q) = \neg p \vee \neg q$

$\neg(p \vee q) = \neg p \wedge \neg q$

Conditional Statements

$p \rightarrow q$

hypothesis — conclusion

antecedent — consequent

$p \vee q \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

$p \rightarrow q \equiv \neg p \vee q$

$\neg(p \rightarrow q) \equiv p \wedge \neg q$

Contrapositive

$p \rightarrow q$ is $\neg q \rightarrow \neg p$

Converse

of $p \rightarrow q$ is $q \rightarrow p$

Inverse

of $p \rightarrow q$ is $\neg p \rightarrow \neg q$

Biconditional

$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

order of operations for Logical operators

1) $\neg$
2) $\wedge, \vee$
3) $\rightarrow, \leftrightarrow$

Theorem 2.1.1 Logical Equivalences

Given any statement variables p, q, and r, a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

1.	*Commutative laws:*	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
2.	*Associative laws:*	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
3.	*Distributive laws:*	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
4.	*Identity laws:*	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
5.	*Negation laws:*	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
6.	*Double negative law:*	$\sim(\sim p) \equiv p$	
7.	*Idempotent laws:*	$p \wedge p \equiv p$	$p \vee p \equiv p$
8.	*Universal bound laws:*	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
9.	*De Morgan's laws:*	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
10.	*Absorption laws:*	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
11.	*Negations of* $\mathbf{t}$ *and* $\mathbf{c}$*:*	$\sim\mathbf{t} \equiv \mathbf{c}$	$\sim\mathbf{c} \equiv \mathbf{t}$

Modus Ponens	$p \rightarrow q$ p $\therefore q$	**Elimination**	**a.** $p \vee q$, $\sim q$, $\therefore p$ **b.** $p \vee q$, $\sim p$, $\therefore q$
Modus Tollens	$p \rightarrow q$ $\sim q$ $\therefore \sim p$	**Transitivity**	$p \rightarrow q$ $q \rightarrow r$ $\therefore p \rightarrow r$
Generalization	**a.** p, $\therefore p \vee q$ **b.** q, $\therefore p \vee q$	**Proof by Division into Cases**	$p \vee q$ $p \rightarrow r$ $q \rightarrow r$ $\therefore r$
Specialization	**a.** $p \wedge q$, $\therefore p$ **b.** $p \wedge q$, $\therefore q$		
Conjunction	p q $\therefore p \wedge q$	**Contradiction Rule**	$\sim p \rightarrow \mathbf{c}$ $\therefore p$

sufficient condition

$r \rightarrow s$

$\neg r \rightarrow \neg s$

Necessary condition

Sheffer Stroke

$P \mid Q \equiv \neg(P \wedge Q)$

Peirce Arrow

$P \downarrow Q \equiv \neg(P \vee Q)$